

CSE 4111

Compiler Design

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Symbol Table

- Symbol table is an important data structure created and maintained by compilers in order to store information about the occurrence of various entities such as variable names, function names, objects, classes, interfaces, etc. Symbol table is used by both the analysis and the synthesis parts of a c

	<i>key</i>	<i>value</i>
<i>phone book</i>	name	phone number
<i>dictionary</i>	word	definition
<i>account</i>	account number	balance
<i>genomics</i>	codon	amino acid
<i>data</i>	data/time	results
<i>Java compiler</i>	variable name	memory location
<i>file share</i>	song name	machine
<i>Internet DNS</i>	website	IP address

Purposes of using symbol table

A symbol table may serve the following purposes depending upon the language in hand:

- To store the names of all entities in a structured form at one place.
- To verify if a variable has been declared.
- To implement type checking, by verifying assignments and expressions in the source code are semantically correct.
- It is used by the compiler to achieve compile-time efficiency.

What is symbol table? What are the purposes of using symbol table?

Information/Attributes of symbol table

V. Name	Type	Size	Dimension	Line of declaration	Line of usage	Address
x	int	4	0	3	6→10	----
a	char	5	1	4	-----	----

```
main()
{
  Int x;
  char a[5];
  .
  .
  X
  .
}
```

What are the attributes used in symbol table? Describe with examples.

Implementation of Symbol table

Following are commonly used data structures for implementing symbol table:-

1. Linked List
2. Hash Table
3. Binary Search Tree

Linked List –

- This implementation is using a linked list. A link field is added to each record.
- Searching of names is done in order pointed by the link of the link field.
- A pointer “**First**” is maintained to point to the first record of the symbol table.
- Insertion is fast $O(1)$, but lookup is slow for large tables – $O(n)$ on average.

Describe linked list and hash table to implement symbol table.

Implementation of Symbol table (Contd.)

- Hash Table –

- In hashing scheme, two tables are maintained – a hash table and symbol table are the most commonly used method to implement symbol tables.
- A hash table is an array with an index range: 0 to table size – 1. These entries are pointers pointing to the names of the symbol table.
- To search for a name we use a hash function that will result in an integer between 0 to table size – 1.
- Insertion and lookup can be made very fast – $O(1)$.
- The advantage is quick to search is possible and the disadvantage is that hashing is complicated to implement.

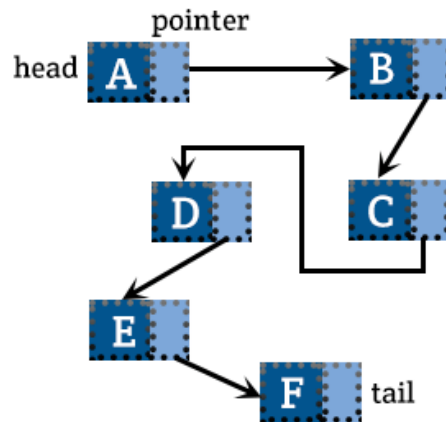
Describe linked list and hash table to implement symbol table.

Implementation of Symbol table (Contd.)

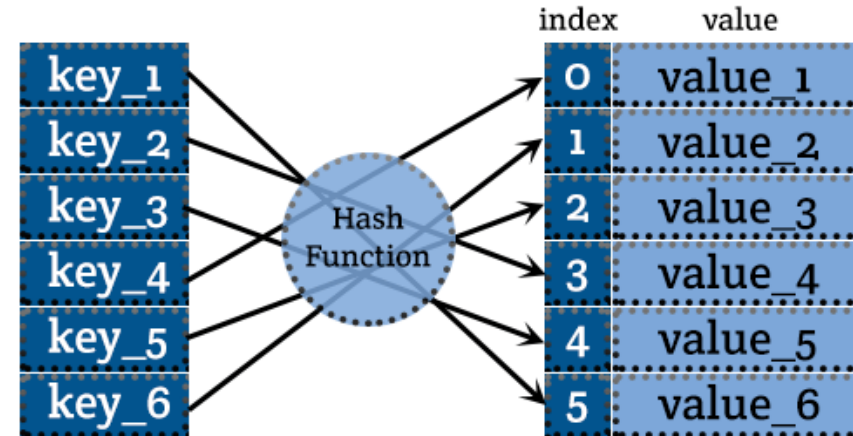
Array

index	
0	A
1	B
2	C
3	D
4	E
5	F

Linked List



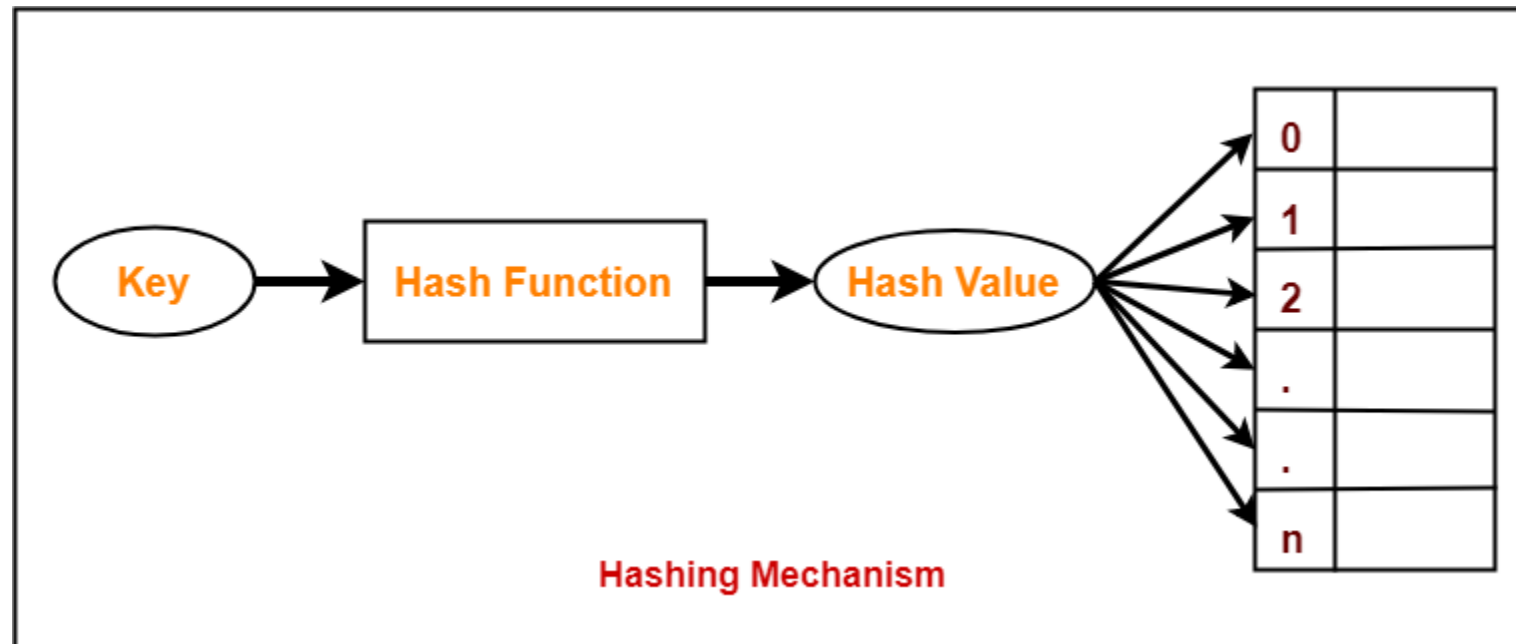
Hash Table



Describe linked list and hash table to implement symbol table.

Hashing

Hashing is the process of transforming any given key or a string of characters into another value. This is usually represented by a shorter, fixed-length value or key that represents and makes it easier to find or employ the original string.



What is hashing? Describe linked list and hash table to implement symbol table.

Hash Functions

Types of Hash functions

There are many hash functions that use numeric or alphanumeric keys. This article focuses on discussing different hash functions:

- **Division Method**
- **Mid Square Method**
- **Folding Method**
- **Multiplication Method**

Division Method:

This is the most simple and easiest method to generate a hash value. The hash function divides the value k by M and then uses the remainder obtained.

Formula:

$$h(K) = k \bmod M$$

Here,

k is the key value, and

M is the size of the hash table.

It is best suited that M is a prime number as that can make sure the keys are more uniformly distributed. The hash function is dependent upon the remainder of a division.

Example

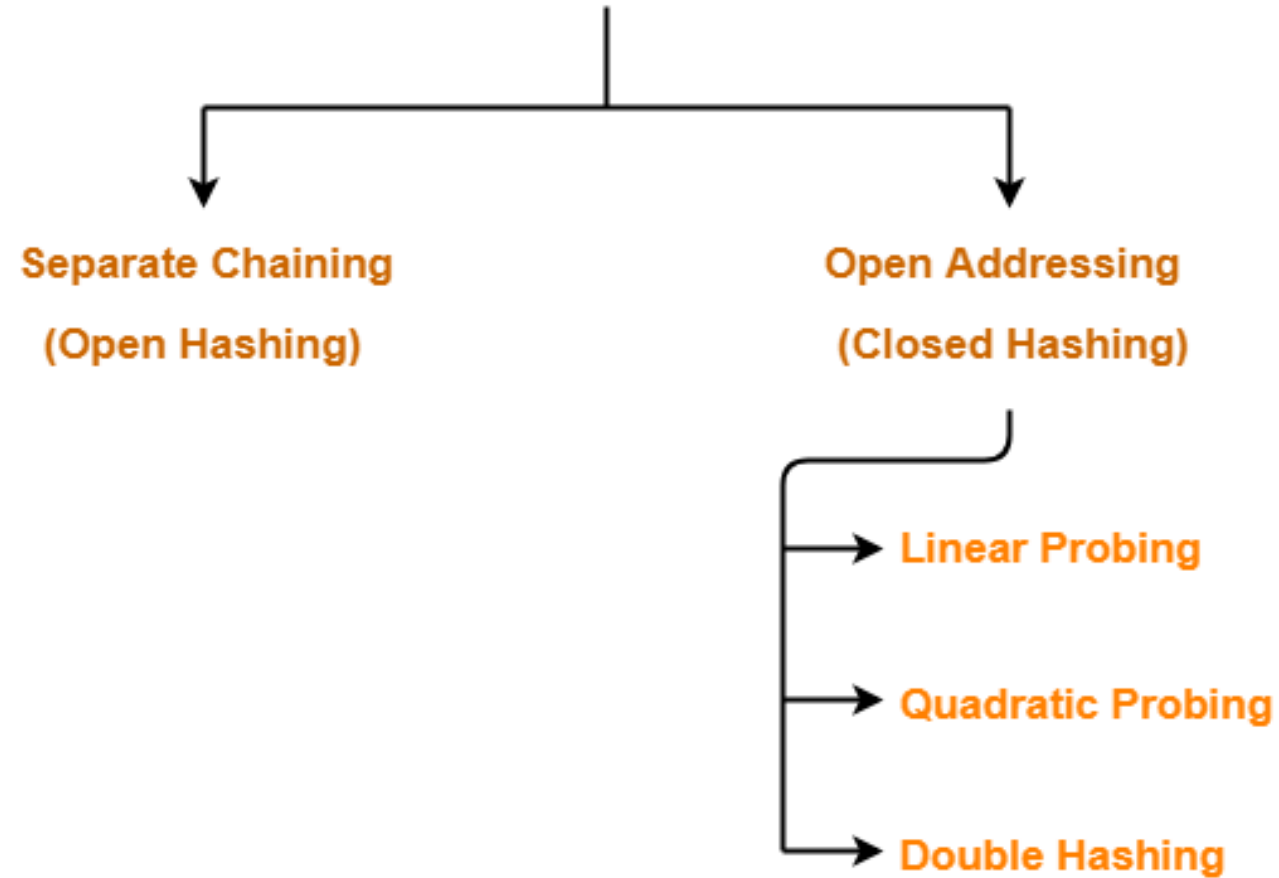
- Search key (24, 52, 91, 67, 48, 83)
- Hash function ($K \bmod 10$)
- If one more value 62 will arrive, then what will be the location? (collision)

	0
91	1
52	2
83	3
24	4
	5
	6
67	7
48	8
	9

Practice this type example.

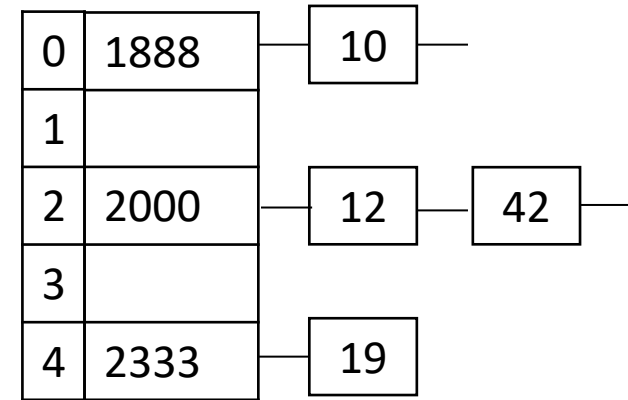
Collision in Hashing

Collision Resolution Techniques



Chaining (Open Hashing)

- $(K \bmod 5)$ function
- Keys (12, 19, 10, 42)
- Here $n=5$ ($k \bmod n$), So the number of locations are 5 (0-4).



Practice this type example.

Chaining (Open Hashing)

Advantages

- Insertion and deletion is easy
- The division method is very fast since it requires only a single division operation.

Disadvantages

- Time complexity $O(n)$
- Waste available spaces

Write down the advantages and disadvantages of open hashing?

Linear Probing

- $R(k) = k \bmod 10$
- $R'(k) = (R(k) + i) \bmod 10$ (collision remover)
- Keys: 43, 135, 72, 23, 99, 19, 82
- i = collision number (no. of checks)
- For 23, $23 \bmod 10 = 3$, $(R(k) + i) = ((3 + 1) \bmod 10) = 4$

19	0
	1
72	2
43	3
23	4
135	5
82	6
	7
	8
99	9

Practice this type example.

Linear Probing

Advantages

- No need of extra spaces

Disadvantages

- Search time complexity $O(n)$
- Deletion difficulty
- Primary clustering
- Secondary clustering

19	0
	1
72	2
43	3
23	4
135	5
82	6
	7
	8
99	9

Write down the advantages and disadvantages of linear probing?

Symbol table representation/organization(Contd.)

Variable-length array

Instead of storing names directly in the symbol table, it will save all names in an independent array of characters and will only keep the start index and length of the name. This will overcome the disadvantage of a fixed-length name.

Ex: int calculate, sum, a, b;

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
c	a	l	c	u	l	a	t	e	\$	s	u	m	\$	a	\$	b	\$	--

Starting index	Length	Type
0	10	int
10	4	int
14	2	int
16	2	int

Difference between fixed length and variable length table representation.

Fixed-length array

Advantages

- It is easy to understand.
- It is easy to implement and access.

Disadvantages

- Wastage of a large amount of memory.
- Sequential access
- Insertion and deletion of records are comparatively very slow.
- It requires large processing and memory requirements.

Write down the advantages and disadvantages of fixed length array representation?

Variable-length array

Advantages

- A variable name can be of any length.
- Ease of Access.
- No wastage of memory in the symbol table.
- The symbol table now more organized.
- It can facilitate quicker editing.

Disadvantages

- The problem of indirection maintenance.
- The requirement of extra array records for storing variable names.

Write down the advantages and disadvantages of variable length array representation?

Operations on Symbol Table

Operations of Symbol table – The basic operations defined on a symbol table include:

Operation	Function
allocate	to allocate a new empty symbol table
free	to remove all entries and free storage of symbol table
lookup	to search for a name and return pointer to its entry
insert	to insert a name in a symbol table and return a pointer to its entry

Insert()

- It is frequently used in analysis phase when token are identified and names are stored in table.
- The insert() function takes the symbol and its value in the form of argument.
- For example, int x; should be processed by compiler as:

Insert(x, int)

Lookup(a)

It is used to search a name & it determine:

- The existence of symbol in the table
- Declaration of the symbol before it is used
- Check whether the name is used in the scope
- Initialization of the symbol
- Checking whether the name is declared multiple times

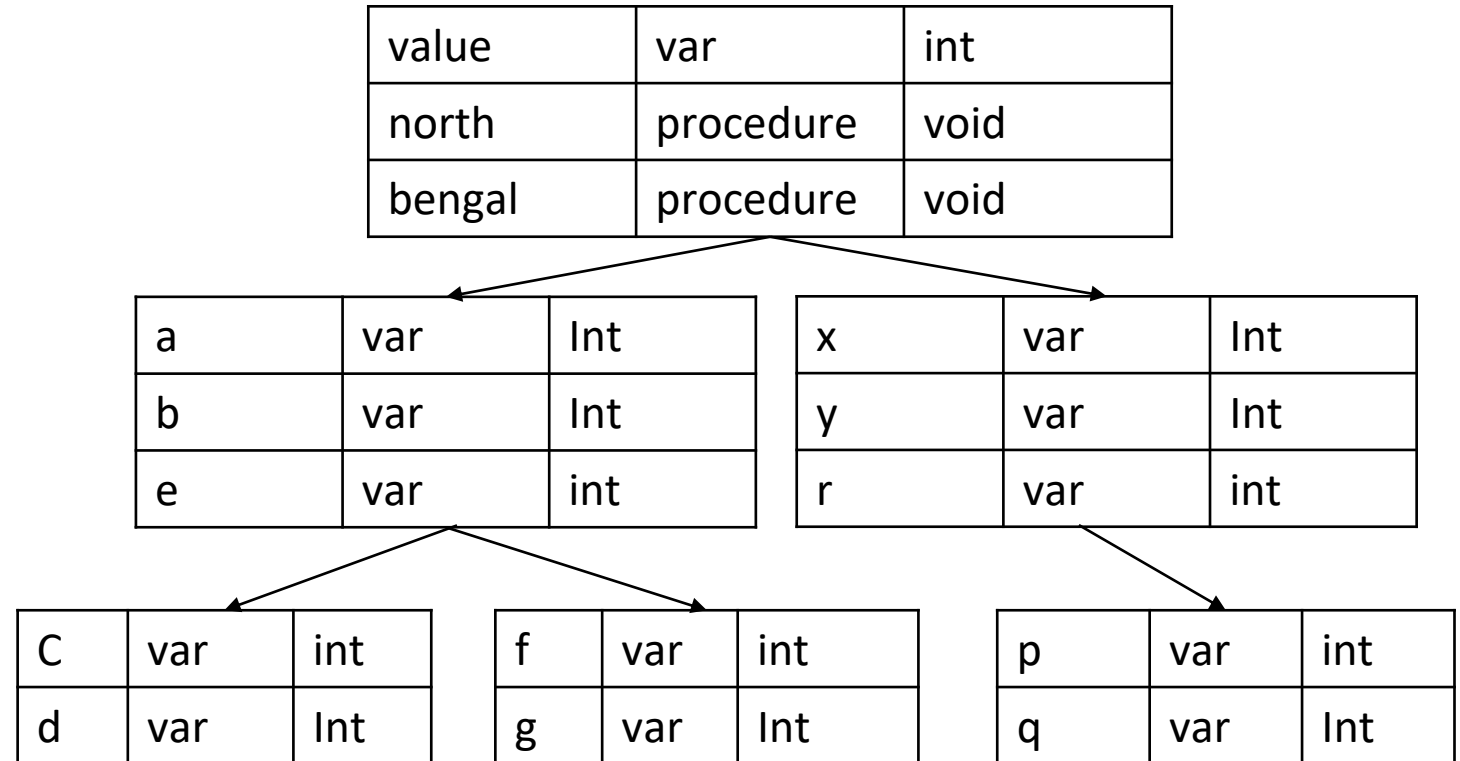
What are the operations we can perform on symbol table. Explain briefly.

Scope Management (Local and Global Variable)

```
Int value;  
void north()  
{  
  Int a;  
  Int b;  
  {  
    int c;  
    Int d;  
  }  
  Int e;  
  {  
    int f;  
    Int g;  
  }  
}
```

```
void bengal()  
{  
  Int x;  
  Int y;  
  {  
    int p;  
    Int q;  
  }  
  Int r;  
}
```

Symbol Table



Describe scope management in symbol table with necessary example.